

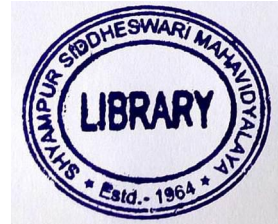
2022

COMPUTER SCIENCE — GENERAL

Paper : GE/CC-2

(Algorithm and Data Structure)

Full Marks : 50



The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **Question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions from the following :

2×5

- (a) Differentiate between Linear and Non-Linear data structure.
- (b) Define a Binary Search Tree.
- (c) What are the advantages of Linked List over Array?
- (d) What is the postfix expression of the given infix expression $(A + B) / (X - Y) * C$?
- (e) Differentiate between Linear Queue and Circular Queue.
- (f) What are the advantages of Binary Search over Linear Search?
- (g) What do you understand by column major representation of a two-dimensional array?
- (h) What do you understand by leaf nodes and internal nodes in a tree?

2. (a) Write an algorithm to insert an element at the beginning and end of singly linked list.

(b) What are the characteristics of an algorithm?

(4+3)=3

3. (a) Define stack.

(b) Write algorithms for the following stack operations, for array implementation of stack :

- (i) Test whether the stack is empty.
- (ii) Test whether the stack is full.
- (iii) Push an element to the stack.
- (iv) Pop an element from the stack.
- (v) Peek the stack.

2+(1+1+2+2+2)

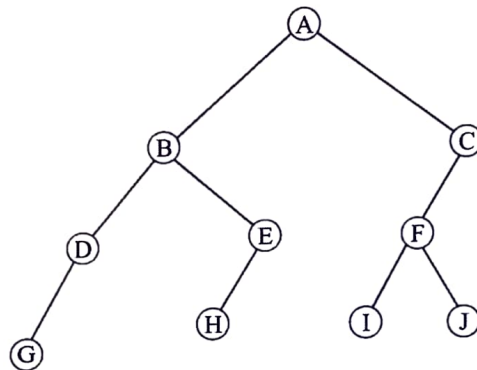
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4. (a) Evaluate the following postfix expression using stack :
8, 2, 3, +, -, 9, 3, /, *.

4+6

- (b) Write algorithm for merge sort.

5. (a) Perform pre-order, in-order and post-order traversal of the following binary tree.



- (b) Write an algorithm to search an element from an array using binary search.

(2+2+2)+4

6. (a) Define BST.

- (b) Construct a BST using the following nodes –

50, 30, 80, 100, 20, 70, 40, 90, 10, 60

Show all the steps in your construction.

- (c) What do you observe when you perform in-order traversal of a BST? Does this apply to the BST above?

2+5+(1+2)

7. (a) Suppose a 2D-array A is declared using A (2:8, 1:4). How many elements can you store in A? What will be the location of A [5] [6] if you store the elements using row-major order (consider w = 4)?

- (b) Write an algorithm for Insertion sort.

(2+3)+5

8. (a) "Binary Search is not possible in linked list" — Justify.

- (b) What are the advantages of doubly linked list over singly linked list?

- (c) Write an algorithm to implement delete at end operation on doubly linked list.

3+3+4